

Chapter 5 *Discrete choice random utility models*

Based on Regis Renault's Slides

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- ① Background on product differentiation.
- ② Linear random utility model of demand.
- ③ Multinomial logit
- ④ Oligopoly and price competition.
- ⑤ Mixed logit and random coefficients.

1 Background on product differentiation

Horizontal versus vertical differentiation

- Product's attributes often thought of as summarized in its quality (e.g. a better computer has a faster chip, more memory and better video).
- Then consumers would agree on which product is better.
- However consumer tastes are heterogeneous: some might want a big high performance laptop, while others would prefer a small and light one.
- The theory of imperfect competition with product differentiation distinguishes
 - Vertical product differentiation: with equal prices all consumers would prefer one product to the other.
 - Horizontal differentiation: at equal prices, different consumers would select different products.

1 Background on product differentiation

Horizontal versus vertical differentiation

Three remarks

- The two types of differentiation have different implications for pricing, both in monopoly and oligopoly..
- There can be taste heterogeneity with vertical differentiation (e.g. the Mussa Rosen setting).
- Empirical challenge in identifying and measuring product quality.

1 Background on product differentiation

Localized versus non localized competition

- Product choice is an essential dimension of non price competition (Hotelling, 1929).
- With more than two products each firm might compete only with a subset of other firms (localized competition, e.g. the Vickrey/Salop circle model where each competes with its two neighbors).
- With enough dimensions in the product space, a firm can compete with all others (non localized competition).

1 Background on product differentiation

Discrete choice versus multi homing

- Discrete choice means that each consumer consumes only one of the products.
- A consumer could conceivably purchase multiple products from multiple sellers (situation described as “multi homing” in the platform literature).
- If this is the case then there is some complementarity between products which softens competition.
- Early models of non localized competition assumed a representative consumer that consumes all varieties, rather than discrete choice (Spence, Dixit, Stiglitz). This approach has not been pursued in IO (rather in Trade, macro and growth theory).

2 Linear random utility model of demand

- For each consumer $\ell = 1, \dots, L$ there are n draws $\epsilon_{\ell,i}$, $i = 1, \dots, n$, where $\epsilon_{\ell,i}$ are i.i.d across consumers ℓ and $E\epsilon_{\ell,i} = 0$ for all ℓ and i .
- Support of $(\epsilon_{\ell,1}, \dots, \epsilon_{\ell,n})$ is $[a, b]^n$, density is f .
- Consumer ℓ 's utility if she buys product i at price p_i is

$$u_{\ell,i}(p_i) = \alpha_i - p_i + y_\ell + \epsilon_{\ell,i}. \quad (1)$$

where y_ℓ is ℓ 's income and α_i is a mean valuation for product i (it could for instance depend on some vector of product characteristics X_i with $\alpha_i = X_i\beta$).

- Let $v_i = \alpha_i - p_i$.

2 Linear random utility model of demand

- ℓ prefers product i to product j if and only if

$$v_i + \epsilon_{\ell,i} > v_j + \epsilon_{\ell,j}. \quad (2)$$

- Consumer ℓ prefers not to buy if

$$\max_{i=1,\dots,n} \{v_i + \epsilon_{\ell,i}\} < 0. \quad (3)$$

- No income effect.
- $\alpha_i + \epsilon_{\ell,i}$ can be interpreted as a measure of the match quality between consumer ℓ and product i .

2 Linear random utility model of demand

Determination of demand for product i .

- Consider product 1.
- Consumer ℓ buys product 1 if

$$\epsilon_1 > -v_1 \tag{4}$$

and

$$\epsilon_j < \epsilon_1 + v_1 - v_j. \tag{5}$$

2 Linear random utility model of demand

Determination of demand for product i .

Expected demand for product 1 can therefore be written as

$$D_1(p_1, \dots, p_n) = L \int_{-v_1}^b \int_a^{\epsilon_1 + v_1 - v_2} \dots \int_a^{\epsilon_1 + v_1 - v_n} f(\epsilon_1, \dots, \epsilon_n) d\epsilon_n, \dots, d\epsilon_1 \quad (6)$$

The joint distribution could be normal

$(\epsilon_1, \dots, \epsilon_n) \sim N((0, \dots, 0), \Sigma)$, which yields a multinomial probit model.

2 Linear random utility model of demand

Two products

With two products product 1's demand is

$$D_1(p_1, p_2) = L \int_{-v_1}^b \int_a^{\epsilon_1 + v_1 - v_2} f(\epsilon_1, \epsilon_2) d\epsilon_2 d\epsilon_1. \quad (7)$$

The set over which the integral is calculated can be depicted graphically.

This shows how changes in prices affect product 1's market share.

2 Linear random utility model of demand

Covered market

Because of budget constraints the maximum relevant price is $\bar{p} = \max\{y_1, \dots, y_n\}$ (none of the products could be purchased by any consumer at larger prices)

Now assume that $a + \min_i\{\alpha_i\} > \bar{p}$, then for any product i , $-v_i = p_i - \alpha_i < a$.

Then all consumers buy one product with probability one and product 1's demand is.

$$D_1(p_1, \dots, p_n) = L \int_a^b \int_a^{\epsilon_1 + v_1 - v_2} \dots \int_a^{\epsilon_1 + v_1 - v_2} f(\epsilon_1, \dots, \epsilon_n) d\epsilon_n, \dots, d\epsilon_1. \quad (8)$$

Specification is only useful to buy some tractability in the theory (especially when combined with symmetry or duopoly).

2 Linear random utility model of demand

i.i.d. matches across products

Assume now that $\epsilon_{\ell,i}$, $\ell = 1, \dots, L$, $i = 1, \dots, n$ are not only i.i.d. across consumers ℓ but also across products.

F and f respectively denote the common c.d.f and density of $\epsilon_{\ell,i}$. Then product 1's demand is.

$$D_1(p_1, \dots, p_n) = L \int_{-v_1}^b \prod_{i=2}^n F(\epsilon_1 + v_1 - v_i) f(\epsilon_1) d\epsilon_1. \quad (9)$$

Much of existing theoretical work uses variants of this version. The multinomial logit is a special case.

2 Linear random utility model of demand

Symmetric products

Special case of previous one where $\alpha_i = \alpha_j = \alpha$ for all $i, j = 1, \dots, n$.

If all products have the same price p^* then all products have the same demand

$$D_1(p^*, \dots, p^*) = L \int_{p^* - \alpha}^b F(\epsilon_1)^{n-1} f(\epsilon_1) d\epsilon_1 = \frac{L}{n} (1 - F(p^* - \alpha)^n) \quad (10)$$

If market is covered, $p^* - \alpha$ replaced by a so each demand is $\frac{L}{n}$.

2 Linear random utility model of demand

Price sensitive individual demand.

Rather than assuming unit demand assume that, if consumer ℓ buys product i , she purchases a quantity $q_{\ell,i}$ and utility is now given by

$$u_{\ell,i}(p_i) = \alpha_i + v(p_i) + y_{\ell,i}, \quad (11)$$

Where $v(p_i) + y_{\ell}$ is the indirect utility from consumer product i at price p_i , which is strictly decreasing and convex in p_i (here, since there is no income effect, it is merely consumer surplus).

Then at price p_i consumer buys $d_i(p_i) = -v'(p_i)$.

Letting $v_i = \alpha_i + v(p_i)$ demand is

$$\begin{aligned} D_1(p_1, \dots, p_n) \\ = L d_1(p_1) \int_{-v_1}^b \int_a^{\epsilon_1 + v_1 - v_2}, \dots, \int_a^{\epsilon_1 + v_1 - v_n} f(\epsilon_1, \dots, \epsilon_n) d\epsilon_n, \dots, d\epsilon_1 \end{aligned}$$

3 Multinomial logit.

- The multinomial logit is a model of probabilistic choice.
- In our context, with $n + 1$ alternatives (including the outside “no purchase” option) the probability that a consumer ℓ chooses product i is

$$P_{\ell,i} = \frac{e^{\frac{v_i}{\mu}}}{\sum_{j=0}^n e^{\frac{v_j}{\mu}}}, \quad (12)$$

where $\mu > 0$ and the outside option is alternative 0 (v_0 remains to be defined).

- μ is a scaling parameter that measures how difference in expected utilities affect choice: it can be construed as a measure of product differentiation.

3 Multinomial logit

The double exponential distribution

- Assume $\epsilon_{\ell,i}$ is i.i.d across consumers and products with c.d.f.

$$\bar{F}(x) = e^{-e^{-\frac{x}{\mu}-\gamma}}, \quad (13)$$

where γ is Euler's constant ($\simeq .5772$).

- ϵ has mean zero, variance $\frac{\mu^2\pi^2}{6}$ and support $(-\infty, +\infty)$.
- Density is

$$f(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}-\gamma} e^{-e^{-\frac{x}{\mu}-\gamma}}. \quad (14)$$

3 Multinomial logit

Covered market

- Relevant demand expression is (9) where the lower bound of the integral $-v_1$ is replaced by a .
- Let us define $t = e^{-\frac{\epsilon_1}{\mu} - \gamma}$ so that $dt = -\frac{1}{\mu} e^{-\frac{\epsilon_1}{\mu} - \gamma} d\epsilon_1$ and $f(\epsilon_1) d\epsilon_1 = -e^{-t} dt$ (we use this to make a change of variable in the integral).
- Also define $Y_i = e^{\frac{v_i}{\mu}}$, $i = 1, \dots, n$ so we have

$$F(\epsilon_1 + v_1 - v_i) = e^{-t \frac{Y_i}{Y_1}}. \quad (15)$$

3 Multinomial logit

Covered market

- Firm 1's demand can be written as

$$D_1(p_1, \dots, p_n) = \int_0^\infty \prod_{i=2}^n e^{-t \frac{Y_i}{Y_1}} e^{-t} dt \quad (16)$$

$$= \int_0^\infty e^{-t \frac{\sum_{i=1}^n Y_i}{Y_1}} dt \quad (17)$$

$$= \left[-\frac{Y_1}{\sum_{i=1}^n Y_i} e^{-t \frac{\sum_{i=1}^n Y_i}{Y_1}} \right]_0^{+\infty} = \frac{e^{\frac{v_1}{\mu}}}{\sum_{i=1}^n e^{\frac{v_i}{\mu}}}. \quad (18)$$

3 Multinomial logit

Outside option

- To have an outside option and keep the multinomial logit form we need to make the associated utility random: assume that not buying any product yields:

$$u_{\ell,0} = \epsilon_{\ell,0} + y_{\ell}, \quad (19)$$

where $\epsilon_{\ell,0}$ is i.i.d across consumers and with all product specific random terms $\epsilon_{\ell,i}$, $i = 1, \dots, n$.

- Then consumer ℓ chooses alternative $i = 0, \dots, n$ with probability given by (12) with $v_0 = 0$, that is

$$P_{\ell,i} = \frac{e^{\frac{v_i}{\mu}}}{1 + \sum_{j=1}^n e^{\frac{v_j}{\mu}}}, \quad (20)$$

3 Multinomial logit

Econometric specification

- Assume $\alpha_i = x_i \bar{\beta}$, where x_i is a vector of characteristics for product i .
- Then we can write

$$\frac{v_i}{\mu} = x_i \beta - \delta p_i, \quad (21)$$

where $\beta = \frac{1}{\mu} \bar{\beta}$ and $\delta = \frac{1}{\mu}$.

- Then individual demand can be estimated using a standard multinomial logit
- Under the assumption that $\epsilon_{\ell,i}$ is i.i.d across consumers, aggregate demand for product i is merely $D_i = LP_{\ell,i}$.

3 Multinomial logit

IIA property

- Take a consumer ℓ who consumes either product 1 or product 2: then the conditional probability that 1 is chosen is

$$\frac{P_{\ell,1}}{P_{\ell,1} + P_{\ell,2}} = \frac{e^{\frac{v_1}{\mu}}}{e^{\frac{v_1}{\mu}} + e^{\frac{v_2}{\mu}}}. \quad (22)$$

It does not depend on the number of other products available or on their characteristics (v_j for $j \neq 1, 2$).

- This property is called “independence of irrelevant alternatives” (IIA)..

3 Multinomial logit

IIA property

- This is an unappealing property: suppose the choice probabilities for 3 different cars are
 - 1 Renault Clio: .5
 - 2 Mercedes S class: .25
 - 3 Porsche Cayenne: .25
- IIA says that if Porsche Cayenne is taken out of the market, probabilities for Renault Clio and Mercedes S Class go up to 2/3 and 1/3 respectively (which were the conditional probabilities when the Porsche was still around).
- At the aggregate level, the market share for the Renault would increase more than that of the Mercedes.
- For a price change we have $\frac{dP_{\ell,i}}{dp_3} = \frac{1}{\mu} P_{\ell,i} P_{\ell,3}$, $i = 1, 2$, so the brand with a large share is more affected by a change in price of the third brand.

4 Oligopoly

- Assume now that each product i is produced at constant marg. cost c_i .
- Products are sold by $M \leq n$ firms.
- Assuming $\epsilon_{\ell,i}$ i.i.d over ℓ and i , $i = 0, \dots, n$ demand for product i is

$$D_i = L \int_a^b \Pi_{j \neq i} F(\epsilon + v_i - v_j) f(\epsilon) d\epsilon, \quad (23)$$

with $v_0 = 0$.

- Firms choose prices simultaneously in a Nash equilibrium.
- p_i^* : equilibrium price of product i .

4 Oligopoly

First order conditions: single product firms

- Assume now each firm sells one product so $M = n$ (and we call i the firm selling product i).
- Consider firm i 's problem: it selects a price p_i expecting other firms to charge equilibrium prices p_j^* , $j \neq i$.
- Its demand deriv. with respect to own price p_i is

$$\frac{\partial D_i}{\partial p_i} = -L \int_a^b \sum_{j \neq i} f(\epsilon + v_i - v_j) \Pi_{k \neq i, j} F(\epsilon + v_i - v_k) f(\epsilon) d\epsilon. \quad (24)$$

4 Oligopoly

First order conditions: single product firms

- As for a single product monopolist the price FOC is

$$p_i^* - c_i = -\frac{D_i}{\partial D_i / \partial p_i}. \quad (25)$$

- Jointly estimating demand and this FOC would yield estimates for demand parameters and marginal costs.
- However actual data concern multi product firms so the price FOCs will be more complex.

4 Oligopoly

First order conditions: multi product firms

- Now assume each firm m sells several products and denote K_m the set of products sold by firm m .
- Firm m 's profit can be written as

$$\pi_m = \sum_{i \in K_m} (p_i - c_i) D_i. \quad (26)$$

- Hence, FOC for p_i is

$$p_i^* - c_i = -\frac{D_i}{\partial D_i / \partial p_i} - \sum_{j \in K_m, j \neq i} \frac{\partial D_j / \partial p_i}{\partial D_i / \partial p_i} (p_j^* - c_j). \quad (27)$$

4 Mixed logit and random coefficients

Consumer heterogeneity in the LRUM model

- In the LRUM model, consumer heterogeneity enters in two ways:
 - ① through income y_ℓ .
 - ② through the random term ϵ_ℓ .
- We noted already that income plays no role because it cancels out in the choice.
 - if we introduce a vector of other individual characteristics z_ℓ that enters additively in utility as $z_\ell \gamma$ it would also be canceled out.

4 Mixed logit and random coefficients

Consumer heterogeneity in the LRUM model

- The random term $\epsilon_{\ell,i}$ does not interact with product specific variables summarized in v_i .
- As a result, market shares are fully determined by the values of v_i independent of its composition:
 - e.g. demand for a high quality (high α_i) product with a high price could be analogous to that of a low quality product with a low price (same market share, same price derivatives).
 - observed product characteristics only matter through their contribution to α_i but it does not matter whether two products have different or similar characteristics.
- IIA for the logit is an extreme consequence of this.

4 Mixed logit and random coefficients

Mixed logit

- When dealing with individual choice data, individual heterogeneity and individual characteristics can be accounted for through a *mixed logit* model.
- In our econometric specification of the logit model, we could for instance assume that the price parameter is consumer specific and can be written $\delta_\ell = z_\ell \gamma + \eta_\ell$ where η_ℓ is i.i.d across consumers.
- A similar strategy could be used for the parameters in β associated with the various product characteristics.
- Then choice probabilities are obtained by taking the mean of the logit expression over the random terms.

4 Mixed logit and random coefficients

Aggregate demand data and random coefficients

- With aggregate demand data, random coefficients can be introduced with aggregate data on the distribution of individual characteristics (e./g. distribution of income).
- Then the market share is computed as the mean over all realizations of z_ℓ of the mixed logit probabilities.